PROBLEM 48

# Two Connected Particles, One on Paraboloid

Original statement · Problems.pdf, p. 33

Multi-Particle Manifold

Two heavy particles $P$ and $Q$, each of mass $m$, are connected with an inextensible, massless string of length $\ell$ as shown in the figure. Particle $P$ is constrained to stay on a paraboloid, given implicitly by equation $z = cR^2$, where $c$ is a known constant and we have introduced the usual cylindrical coordinates $\{R, \theta, z\}$. The string passes through a small hole at the origin $O$, and particle $Q$ is restricted to hang along the vertical axis. The reference frame $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial. There is no friction in this problem. Assume that the string is long enough so that $Q$ never reaches $O$ and that it is always tense.

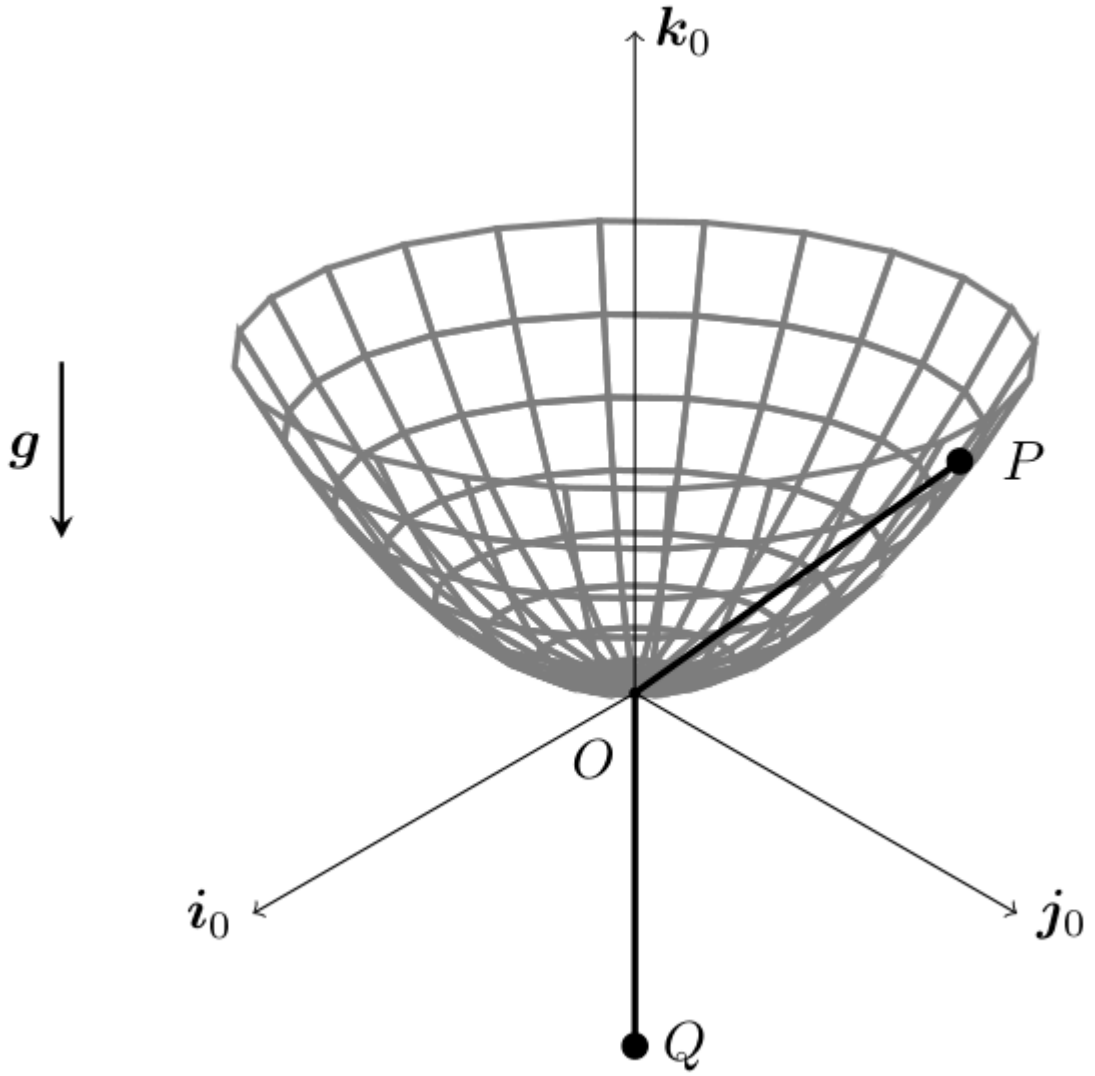



*Figure: Paraboloid with P on its surface, string through the hole at O and Q hanging along the vertical axis k0 (Problems.pdf, p. 33).*

**(a)** How many effective degrees of freedom does the system have? justify your answer.

**(b)** Give the expressions for the position, velocity, and acceleration of $P$ and $Q$ with respect to $S_0$ as a function of $R, \theta$ and their derivatives. If you use any vector basis other than $\{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ to write your solution, define it first.

**(c)** Find a normal vector and two linearly independent tangent vectors to the paraboloid as a function of $R, \theta$. They do not need to be unitary.

**(d)** Identify and write down the general expression of all the forces acting on $P$ and $Q$, introducing new unknowns as needed.

**(e)** Find the condition that the solution $R(t)$ must satisfy so that the string is indeed always in tension.

**(f)** Using $S_0$ for your analysis, write down two second-order differential equations for $R$ and $\theta$, that do not depend on any other unknowns.

**(g)** Find two scalar conservation laws (i.e., first integrals).

**(h)** Assuming that initially $R = 1/c, \theta = 0, \dot{R} = 0, \dot{\theta} = d$, where $d \neq 0$ is a given value, write down an equation in the form $f(R) = 0$ whose roots determine the minimum and maximum values of $R$ in the subsequent motion. You do not need to solve this equation.

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**Worked Solution (4-Phase Methodology)**

`PHASE 1`

### Constraints, Degrees of Freedom & Coordinates

- **System:** two heavy particles $P$ and $Q$ of mass $m$ are joined by an inextensible massless cord of length $\ell$ that passes through a small hole at the origin $O$. $P$ moves without friction on the paraboloid $z = cR^2$ ($c > 0$, cylindrical coordinates $\{R, \theta, z\}$); $Q$ is restricted to hang along the vertical axis $Oz$. The frame $S_0$ is inertial, there is no friction, the cord is always taut and $Q$ never reaches $O$.
- **Constraints (a):** two free particles have 6 coordinates, and there are 4 independent constraints:
  - $P$ on the paraboloid: $z_P = cR^2$ (1; bilateral, since the statement says $P$ is constrained to stay on the paraboloid).
  - $Q$ on the axis: $x_Q = y_Q = 0$ (2).
  - Constant cord length: $|\overrightarrow{OP}| + |z_Q| = \ell$ (1; unilateral, $T \geq 0$).
- **Degrees of freedom:** $\mathrm{CDOF} = 6 - 4 = 2$, with $(R, \theta)$.

- **Initial state (h):** $R(0) = 1/c, \theta(0) = 0, \dot{R}(0) = 0, \dot{\theta}(0) = d \neq 0$.
- **Parameters:** $m$, $c$ [$\mathrm{m}^{-1}$], $\ell$, $g$, $d$ [$\mathrm{s}^{-1}$].

PHASE 2

### Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Cylindrical basis:** $\mathcal{B}_1 = \{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{k}\}$ with $\mathbf{k} = \mathbf{k}_0$, the inertial basis rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

with $\dot{\mathbf{e}}_R = \dot{\theta}\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot{\theta}\,\mathbf{e}_R$.

- **Surface-adapted basis.** With $s = \sqrt{1 + 4c^2R^2}$, the unit meridian tangent is $\mathbf{e}_g = (\mathbf{e}_R + 2cR\,\mathbf{k})/s$, the unit normal pointing into the bowl is $\hat{\mathbf{n}} = (-2cR\,\mathbf{e}_R + \mathbf{k})/s$, and $\mathbf{e}_g \times \mathbf{e}_\theta = \hat{\mathbf{n}}$, so

$$[{}_1R_2] = \begin{pmatrix} 1/s & 0 & -2cR/s \\ 0 & 1 & 0 \\ 2cR/s & 0 & 1/s \end{pmatrix}, \qquad \det[{}_1R_2] = \frac{1 + 4c^2R^2}{s^2} = 1, \qquad [{}_1R_2][{}_1R_2]^T = I$$

PHASE 3

### Derivation

#### 3.1 Kinematics

**(b) Position, velocity and acceleration**

**Pedagogical Justification:** the cord runs straight from $O$ to $P$, so the length of its part on the side of $P$ is $\rho(R) = |\mathbf{r}_P| = R\sqrt{1 + c^2R^2}$, and $Q$ hangs at the depth $\ell - \rho$:

$$\mathbf{r}_P = R\,\mathbf{e}_R + cR^2\,\mathbf{k}, \qquad \mathbf{r}_Q = -\left(\ell - \rho(R)\right)\mathbf{k}$$

- **Derivatives of $\rho$** (chain rule, with $w = \sqrt{1 + c^2R^2}$): $\rho' = \dfrac{d\rho}{dR} = w + \dfrac{c^2R^2}{w} = \dfrac{1 + 2c^2R^2}{w}$ and $\rho'' = \dfrac{c^2R\left(3 + 2c^2R^2\right)}{w^3}$.
- **Particle $P$** (product rule and Poisson):

$$\mathbf{v}_P = \dot{R}\,\mathbf{e}_R + R\dot{\theta}\,\mathbf{e}_\theta + 2cR\dot{R}\,\mathbf{k}, \qquad v_P^2 = \dot{R}^2\left(1 + 4c^2R^2\right) + R^2\dot{\theta}^2$$

$$\mathbf{a}_P = \left(\ddot{R} - R\dot{\theta}^2\right)\mathbf{e}_R + \left(R\ddot{\theta} + 2\dot{R}\dot{\theta}\right)\mathbf{e}_\theta + 2c\left(\dot{R}^2 + R\ddot{R}\right)\mathbf{k}$$

- **Particle $Q$:** $\mathbf{v}_Q = \rho'\dot{R}\,\mathbf{k}$ and $\mathbf{a}_Q = \left(\rho'\ddot{R} + \rho''\dot{R}^2\right)\mathbf{k}$.

**(c) Tangent and normal vectors to the paraboloid**

$$\mathbf{t}_R = \frac{\partial \mathbf{r}_P}{\partial R} = \mathbf{e}_R + 2cR\,\mathbf{k}, \qquad \mathbf{t}_\theta = \frac{\partial \mathbf{r}_P}{\partial \theta} = R\,\mathbf{e}_\theta, \qquad \mathbf{n} = \mathbf{t}_R \times \mathbf{t}_\theta = R\left(-2cR\,\mathbf{e}_R + \mathbf{k}\right)$$

### 3.2 Forces

**(d) Forces on $P$ and $Q$**

- **On $P$:** its weight $-mg\,\mathbf{k}$; the reaction of the smooth paraboloid $\mathbf{N} = N\,\hat{\mathbf{n}}$ (unknown magnitude $N$, normal to the surface); and the tension of the cord, directed toward the hole, $\mathbf{T}_P = -T\,\dfrac{\mathbf{r}_P}{|\mathbf{r}_P|}$.
- **On $Q$:** its weight $-mg\,\mathbf{k}$, the tension $\mathbf{T}_Q = +T\,\mathbf{k}$ (same cord, smooth hole), and the horizontal reaction $\mathbf{R}_Q$ of the vertical guide.
- **Inertia forces:** none ($S_0$ is inertial).

### 3.3 Equations of Motion

**(f) Two second-order equations for $R$ and $\theta$**

> **Pedagogical Justification:** the unknowns are $R, \theta, T, N$ and the two components of $\mathbf{R}_Q$, i.e. 6 unknowns, and Newton's second law for $P$ and $Q$ gives 6 scalar equations. Projecting the equation of $P$ on the tangent vectors removes the normal reaction $N$, and the equation of $Q$ gives $T$ and $\mathbf{R}_Q = \mathbf{0}$ (since $\mathbf{a}_Q \parallel \mathbf{k}$):

$$Q,\ \mathbf{k}:\ m\left(\rho'\ddot{R} + \rho''\dot{R}^2\right) = T - mg \implies T = m\left(g + \rho'\ddot{R} + \rho''\dot{R}^2\right)$$

- **Equation for $\theta$.** Projecting on $\mathbf{e}_\theta$ (no component of the reaction, of gravity or of the tension, which lies in the meridian plane): $m\left(R\ddot{\theta} + 2\dot{R}\dot{\theta}\right) = 0$, i.e. $\dfrac{m}{R}\dfrac{d}{dt}\left(R^2\dot{\theta}\right) = 0$.
- **Equation for $R$.** Projecting on $\mathbf{t}_R$: the tension component is $-T\,\dfrac{\mathbf{r}_P}{|\mathbf{r}_P|}\cdot\mathbf{t}_R = -T\rho'$ (because $\dfrac{\mathbf{r}_P}{|\mathbf{r}_P|}\cdot\mathbf{t}_R = \dfrac{R + 2c^2R^3}{Rw} = \rho'$) and the weight gives $-mg\,\mathbf{k}\cdot\mathbf{t}_R = -2cmgR$:

$$m\left[\ddot{R} - R\dot{\theta}^2 + 2cR\cdot 2c\left(\dot{R}^2 + R\ddot{R}\right)\right] = -2cmgR - T\rho'$$

Substituting $T$ from the equation of $Q$:

$$\boxed{\left(1 + 4c^2R^2 + \rho'^2\right)\ddot{R} + \left(4c^2R + \rho'\rho''\right)\dot{R}^2 - R\dot{\theta}^2 + 2cgR + g\rho' = 0}, \qquad \boxed{R\ddot{\theta} + 2\dot{R}\dot{\theta} = 0}$$

with $\rho'^2 = \dfrac{(1 + 2c^2R^2)^2}{1 + c^2R^2}$.

### 3.4 Integration and Answers

**(e) Condition for the cord to remain in tension**

> **Pedagogical Justification:** the cord can only pull, so $T > 0$ at all times. From the equation of $Q$:

$$T = m\,(g + \ddot{z}_Q) > 0 \iff \ddot{z}_Q = \rho'\ddot{R} + \rho''\dot{R}^2 > -g, \qquad \text{i.e.} \quad \frac{1 + 2c^2R^2}{\sqrt{1 + c^2R^2}}\ddot{R} + \frac{c^2R\left(3 + 2c^2R^2\right)}{\left(1 + c^2R^2\right)^{3/2}}\dot{R}^2 > -g$$

**(g) Two scalar conservation laws (first integrals)**

- **Azimuthal integral (first integral of the $\theta$ equation).** From the equation for $\theta$ and $a_\theta = \frac{1}{R}\frac{d}{dt}\left(R^2\dot{\theta}\right)$ (product rule):

$$\boxed{R^2\dot{\theta} = \text{const} = R(0)^2\,d = \frac{d}{c^2}} \implies \dot{\theta} = \frac{d}{c^2R^2}$$

- **Mechanical energy.** The reaction of the paraboloid does no work ($\mathbf{N}\cdot\mathbf{v}_P = 0$), the tension does the work $-T\rho'\dot{R}$ on $P$ and $+T\rho'\dot{R}$ on $Q$, which cancel (inextensible cord, smooth hole), the guide reaction is perpendicular to $\mathbf{v}_Q$, and gravity is conservative. With $V = mg\left(cR^2 + z_Q\right) = mg\left[cR^2 + \rho - \ell\right]$:

$$E = \tfrac{1}{2}m\left[\left(1 + 4c^2R^2 + \rho'^2\right)\dot{R}^2 + R^2\dot{\theta}^2\right] + mg\left(cR^2 + \rho\right) - mg\ell = \text{const}$$

**(h) Equation $f(R) = 0$ for the extreme values of $R$**

> **Pedagogical Justification:** at the extremes $\dot{R} = 0$, so only $\frac{1}{2}mR^2\dot{\theta}^2$ remains of the kinetic energy, with $R^2\dot{\theta}^2 = d^2/(c^4R^2)$ from the azimuthal integral. The constant $-mg\ell$ cancels in $E - E_0$. At $R(0) = 1/c$: $\rho = \sqrt{2}/c$ and $R^2\dot{\theta}^2 = d^2/c^2$. Per unit mass:

$$\boxed{f(R) = \frac{d^2}{2c^4R^2} + g\left(cR^2 + R\sqrt{1 + c^2R^2}\right) - \frac{d^2}{2c^2} - \frac{(1 + \sqrt{2})\,g}{c} = 0}$$

- The allowed radii are those with $f(R) \leq 0$ (so that $\dot{R}^2 \geq 0$), and $R = 1/c$ is a root.

PHASE 4

### Interpretation & Checks

- **Bounded precessing motion:** the particle moves between two parallel circles $R_{\min}$ and $R_{\max}$, with the azimuthal angle monotonic ($\dot{\theta} = d/(c^2 R^2)$ has the sign of $d$).
- **Dimensional check:** $[c] = \mathrm{m}^{-1}$, $[d] = \mathrm{s}^{-1}$; $[cR^2] = [\rho] = \mathrm{m}$ and $cR$ is dimensionless; every term of $f$ has units $\mathrm{m}^2/\mathrm{s}^2$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 48 in its solution keys, so the results are verified by independent derivation. Sympy confirms $\rho'$, $\rho''$, the projections of $\mathbf{a}_P$ on $\mathbf{t}_R$ and $\mathbf{e}_\theta$, and numerical integration of the two equations of (f) ($c = d = g = 1$) conserves $E$ to $2 \times 10^{-12}$, keeps $T/m \geq 0.303$ and bounds $R$ between $0.482807$ and $1$, the roots of $f$ ($f(0.482807) = 0$ and $f(1) = 0$; $f < 0$ between them).

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*Source: Problems.pdf*